Precision Voltage Divider

Unity creation guide and import checklist | Editable ScriptableObject included separately

01 Learning objective

Build a 5 V resistor voltage divider using TOP = 330 ohm and BOTTOM = 470 ohm. The output between the resistor midpoint (VOUT) and supply negative should measure approximately 2.9375 V. Current through the series branch should be approximately 6.25 mA.

02 Circuit topology

Supply (+) -> **Power_Switch** -> **Register_Slot_TOP (330 ohm)** -> **VOUT** -> **Register_Slot_BOTTOM (470 ohm)** -> **Supply (-)**.

For measurement, connect multimeter red to VOUT and black to supply negative. Never short VOUT directly to negative using a normal wire.

03 Create the original level assets

• In Assets/My_Assets/_Project_Spark/Prefabs/Levels/, create or open **Level_18_Layout** and its existing **Level_18.asset**. Keep original asset GUID and references.

• Open the existing level SO text importer/exporter. Select the **original Level_18.asset** in Unity and import **Level_18_Voltage_Divider_Editable.txt**. Do not create a new asset from the text file.

• Confirm the Inspector now shows **level_18**, **Precision Voltage Divider**, and 5 targets rather than the old Level 17 LED network.

04 Build the prefab hierarchy

Level_18_Layout (root with SparkLevelLayout and SparkLevelSceneBindings)
  Power_Sources / PowerSupply_01
  Switch / Power_Switch
  Register_Slot_TOP (SparkShelfSlot)
  Register_Slot_BOTTOM (SparkShelfSlot)
  Measurement_Points / VOUT_Point (optional visual marker; add a real probe-compatible terminal only if required by your meter)
  Level_Visuals / input, midpoint and ground labels

05 Configure workstation slots

Field	TOP	BOTTOM
Slot ID	Register_Slot_TOP	Register_Slot_BOTTOM
Category / Kind	Resistor / Workstation	Resistor / Workstation
Accept Any Category	False	False
Enable Smart Binding	True	True
Runtime Component ID	R_TOP_Component	R_BOTTOM_Component
Runtime Terminal A ID	R_TOP_Input	R_BOTTOM_Input

Field	TOP	BOTTOM
Runtime Terminal B ID	R_TOP_Output	R_BOTTOM_Output

06 Declare runtime binding IDs

In the original Level_18_Layout prefab > SparkLevelSceneBindings > Runtime Binding IDs, declare components: **R_TOP_Component, R_BOTTOM_Component**; terminals: **R_TOP_Input, R_TOP_Output, R_BOTTOM_Input, R_BOTTOM_Output**. No R2 aliases or extra electronic-object IDs are needed.

Configure the workstation binding bridge to listen to both slots, map these exact IDs, and perform initial synchronization after registration. Slot occupancy and the bridge must agree.

07 Design the starting inventory

Start with both workstation slots empty, supply off, and available 330 ohm and 470 ohm resistor shelf items. The player installs each item, wires the two resistors in series, powers on, and measures VOUT. If you preinstall items instead, set their actual CurrentSlot and slot Occupant consistently at registration and keep authored transforms.

08 Import the five targets

Target	Type	Component	Range
Output voltage	VoltageRange = 5	R_BOTTOM_Component	2.80–3.10 V
TOP voltage	VoltageRange = 5	R_TOP_Component	1.95–2.20 V
TOP current	CurrentRange = 6	R_TOP_Component	5.8–6.7 mA
BOTTOM current	CurrentRange = 6	R_BOTTOM_Component	5.8–6.7 mA
Source current	CurrentRange = 6	PowerSupply_01	5.8–6.7 mA

Measurement caveat: This asset checks electrical component state; it does not prove the multimeter was used. Voltage sign and terminal ordering must be tested in the actual evaluator; if it compares signed Voltage, reverse the binding terminal order or adapt comparison logic rather than blindly loosening ranges.

09 Validate in Play Mode

• Install 330 ohm in TOP and 470 ohm in BOTTOM; verify CurrentSlot == workstation slot, Occupant == item, and InstalledItem == item.

• Power on and confirm source ~5.00 V, TOP drop ~2.0625 V, BOTTOM drop ~2.9375 V, each series current ~0.00625 A.

• Use diagnostic target logging to verify all five configured target IDs resolve and pass, requireClosedReturn is true, and no short-circuit flag is set.

• Test reversed resistor order, open return, shorted midpoint, removed resistor, supply off, and failed binding; confirm they do not complete the level.

• Inspect physical probe measurement at VOUT and supply negative; do not treat a voltage-target pass as proof of measurement gameplay.

10 Final constraints

The current target ranges approximate the intended pair but do not independently enforce exact resistance values. Implement a resistor-value target if strict value selection is a requirement. The editable file uses your original asset

envelope and existing serialized field names; importer success and in-game completion still require testing in Unity.